

Garv Arora

Robotics & Embedded Systems Engineer | Autonomous Systems | ADAS

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SUMMARY

Computer Science undergraduate (IoT) at VIT promoted to R&D Lead at SEDS VIT — the apex body of SEDS India — after 2+ years building ROS2 autonomous navigation systems, CUDA-accelerated perception pipelines, and ARM bare-metal firmware as an autonomous systems developer. Ranked 12th (IRC 2025) and 16th (IRC 2026) globally among 100+ international teams. Targeting roles in robotics software, embedded systems, ADAS, or autonomous vehicle engineering.

EDUCATION

Vellore Institute of Technology | Vellore, India

CGPA: 8.37 / 10.0

Bachelor of Technology – Computer Science & Engineering (Specialization: IoT)

Gems International School | Gurugram, India

Class 10: 89.3% | Class 12: 76.4%

Central Board of Secondary Education (CBSE)

November 2016 – April 2023

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, SQL, Bash, Verilog, C#, HTML/CSS

Robotics & Autonomous Systems: ROS2, Nav2, MoveIt2, RTAB-Map, Micro-ROS, SLAM, EKF/UKF, PID Control, Point Cloud Processing, mmWave Radar

Perception & ML: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, HuggingFace Transformers

Embedded & Infrastructure: ARM Cortex-M, ESP32, I2C/SPI/UART, FreeRTOS, Docker, Git, GitLab CI/CD, Linux, AWS, PyQt, MongoDB

WORK EXPERIENCE

SEDS Projects VIT | SEDS India

Research & Development Lead

January 2026 – Present

- Promoted to R&D Lead overseeing 2 competition teams (50+ engineers) — Team Ardra (ISDC) and Team Vyadh (IRC/IRDC) — under SEDS India, the national apex body for space engineering students.
- Guided 3 sub-team leads through navigation architecture reviews and hardware integration debugging, resolving 10+ critical pre-competition issues across both teams.

Autonomous Systems Developer – Team Vyadh (Martian Rover)

April 2024 – January 2026

- Architected a full ROS2 navigation stack (Nav2, RTAB-Map Visual SLAM, RealSense D455) generating real-time 3D elevation maps, cutting manual teleoperation time by 70% in competition.
- Slashed CV pipeline latency by ~60% and raised throughput 4x via CUDA, YOLOv8, ArUco/AprilTag detection, and Linux V4L2 tuning across 4 simultaneous camera streams.
- Flashed Micro-ROS firmware to ESP32 microcontrollers for distributed data acquisition; fused IMU and encoder data with an EKF, cutting odometry drift by ~40% and enabling sub-10 cm autonomous positioning.
- Shipped a PyQt5 autonomous mission dashboard with real-time telemetry visualization, parameter tuning, and FSM-based mission sequencing for autonomous competition runs.

Wissen Baum Engineering Solutions

May 2025 – July 2025

Software Automation Intern

Pune, India

- Authored a BDD test suite (Python, Behave, pytest) covering 40+ scenarios, raising regression coverage from 0% to 100% on 3 core product flows in 6 weeks.
- Accelerated GitLab CI/CD by parallelizing 3 stages (Playwright + Cypress), trimming run time ~35% (18 min to 12 min); replaced manual QA with a pandas/NumPy script processing 10,000+ rows per run.
- Enforced code quality across the shared repository by configuring pre-commit Git hooks with linting rules, blocking 100% of non-compliant pushes before merge.

PROJECTS

mmWave Earthquake Survivor Detection Robot | ROS2, Python, Hi-Link LD2410C, ESP32

2026

- Integrated an LD2410C mmWave radar with ROS2 to detect survivor breathing signatures through rubble; fused radar presence data with IMU odometry for autonomous debris-field navigation, marking locations with <15 cm accuracy.
- Streamed radar frames from the LD2410C to an ESP32 over UART, published processed detections as ROS2 topics, and visualised survivor coordinates in real time on a 2D occupancy map.

Gesture-Controlled 5-DOF Robotic Arm | C++, Arduino, MPU6050, Flex Sensors

2026

- Mapped MPU6050 wrist orientation and 2 flex sensor readings to 5-DOF servo commands with <50 ms end-to-end latency; trained a gesture classifier on 200+ samples, achieving 94% accuracy across 8 gestures.
- Transmitted classified gesture commands over an HC-05 Bluetooth link to the arm controller, decoupling the glove unit from the manipulator and enabling wireless operation up to 10 m.

HayaiOS – Bare-Metal RTOS | C, ARM Assembly, ARM Cortex-M4

2026

- Implemented a preemptive round-robin scheduler (1 ms SysTick), context switching in <50 assembly instructions, mutex/semaphore primitives, and a HAL isolating all register-level I/O on ARM Cortex-M4.
- Stress-tested the scheduler under 8 concurrent tasks with simulated interrupt load, confirming zero priority inversion and deterministic <2 μ s context-switch latency across 10,000 scheduling cycles.

CERTIFICATIONS & ACHIEVEMENTS

Oracle Cloud Infrastructure 2025: Generative AI Professional | Data Science Professional

IRC 2025: 12th globally (100+ teams) | **IRC 2026:** 16th globally | **IRDC 2025:** Top 5 Finalist